Exercises 8

Exercise 1 (Quantum correlations). *Let* $U = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ *be an arbitrary unitary operation.*

1. *Let* $|\beta_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. *Compute the state*

$$(U \otimes U^*)|\beta_{00}\rangle$$

where U^* *denotes the entry-wise complex conjugate of* U. *Simplify the result using the fact that* U *is unitary.*

2. *Let* $|\beta_{11}\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$. *Compute the state*

$$(U \otimes U)|\beta_{11}\rangle.$$

How does it compare to the original state $|\beta_{11}\rangle$*?*

3. *How are the measurement outcomes of both qubits correlated if* $(U \otimes U^*)|\beta_{00}\rangle$ *is measured in the standard basis? How about* $(U \otimes U)|\beta_{11}\rangle$*?*

Exercise 2 (Measuring two probabilistic bits). *Let*

$$r = r_{00}[00] + r_{01}[01] + r_{10}[10] + r_{11}[11] = \begin{pmatrix} r_{00} \\ r_{01} \\ r_{10} \\ r_{11} \end{pmatrix}$$

be an arbitrary state on two probabilistic bits, where Alice holds the first bit and Bob holds the second bit.

1. *If Alice measures her bit, what are the probabilities for her to obtain outcomes* 0 *and* 1*, and what are the corresponding post-measurement states of Bob's bit?*

2. *In each case, what is the probability for Bob to obtain the outcome* 0 *if he measures his bit? Average these two probabilities over the probabilities of Alice's measurement outcomes.*

3. *Compare this to the scenario when Bob directly measures the second bit of* r *without waiting for Alice to complete her measurement.*

Exercise 3 (Alice tries to signal to Bob)**.** *Let $|\psi\rangle \in \mathbb{C}^4$ be an arbitrary two-qubit state shared by Alice and Bob. Assume that Alice tries to influence Bob's measurement outcomes by applying a unitary operation U on her qubit.*

1. *What is the probability p_i for Bob to obtain measurement outcome $i \in \{0,1\}$ when measuring the second qubit of $|\psi\rangle$?*
2. *Compute the same probability when measuring the second qubit of $(U \otimes I)|\psi\rangle$. How do they compare?*

Exercise 4 ($H \otimes H$ on Bell states)**.** *Verify that*

$$(H \otimes H)|\beta_{zx}\rangle = (-1)^{zx}|\beta_{xz}\rangle,$$

for all $z, x \in \{0,1\}$, where $|\beta_{zx}\rangle$ are the Bell states.

Hint: *Use the circuit that prepares $|\beta_{zx}\rangle$ from $|z,x\rangle$ and the fact that Hadamarding both inputs and outputs of* CNOT *turns it upside down (recall Exercise 1 from Lecture 5).*

Exercise 5 (CHSH game)**.** *The goal of this exercise is to verify the quantum strategy for the CHSH game.*

1. *Recall the expressions for the following two Bell states:*

$$|\beta_{10}\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle), \qquad |\beta_{01}\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle).$$

For any $\alpha \in \mathbb{R}$, define

$$G(\alpha) = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}$$

and verify that

$$\big(G(\alpha_1) \otimes G(\alpha_2)\big)|\beta_{10}\rangle = \cos(\alpha_1+\alpha_2)|\beta_{10}\rangle + \sin(\alpha_1+\alpha_2)|\beta_{01}\rangle. \tag{1}$$

2. *Assume that both qubits of the state in eq. (1) are measured in the standard basis and denote the two measurement outcomes by $a, b \in \{0,1\}$. What is the probability $p(\alpha_1, \alpha_2)$ that $a \oplus b = 0$ as a function of the two angles α_1 and α_2?*
3. *Let $s, t \in \{0,1\}$ denote the input bits of Alice and Bob, respectively. Recall that Alice's strategy is to apply $G(\theta_s)$ and Bob's strategy is to apply $G(\theta_t)$ on their qubits, where*

$$\theta_0 = -\pi/16, \qquad \theta_1 = 3\pi/16.$$

Verify that their measurement outcomes $a, b \in \{0,1\}$ always satisfy the relation

$$a \oplus b = \mathrm{AND}(s,t)$$

with probability $\cos^2(\pi/8) \approx 0.85$.

Exercise 6 (`Cirq`: Construction of the GHZ-state)**.**

Exercise 7 (`Cirq`: Quantum square game)**.**